

MacroWatch

Development Phases — Phase 1, Phase 2 & Phase 3 Plan

Version: v1.1-stable | Tagged: May 8, 2026 | Live: <https://macro-watch-phi.vercel.app> | GitHub: [mlibrizzi/MacroWatch](#)

Overview: MacroWatch is a macro intelligence dashboard built around Ray Dalio's Big Debt Cycle framework. Development is organized into three phases: Phase 1 (data foundation — complete), Phase 2 (visualization and interpretation — planned), and Phase 3 (investment action layer — future).

PHASE 1 Data Foundation

✓ COMPLETE — v1.1-stable tagged May 8, 2026

Objective

Build a fully functional dashboard with verified real data across all six tabs (Daily, Auctions, Weekly, Monthly, QTR, Intel). Every data point sourced from official APIs with known accuracy and lag characteristics documented.

Completed Fixes

Fix	Description	Files Changed
Fix 1	DXY label corrected to "Fed Broad USD" with subtitle "Broad USD vs 26 currencies (not ICE DXE)"	App.jsx

Fix 2	VIX sourced from FRED VIXCLS (real CBOE data). Added color-coded legend: <15 Complacent 15-20 Normal 20-30 Elevated 30-40 Fear >40 Extreme Fear	fred.js, fetchers.js, App.jsx
Fix 3	Fed Funds Rate hardcoded 3.50%-3.75%. June cut probability hardcoded 6% from CME FedWatch. Eliminates Claude hallucinating wrong rate.	fetchers.js
Fix 4	QTR earnings replaced Claude estimates with real Finnhub/SEC data. AAPL \$2.01, MSFT \$4.27 verified against SEC EDGAR.	fetchers.js, earnings.js
Fix 4b	Government reports anchored to real values: GDP +2.0%, Deficit \$1.307T, Debt \$38.9T, Trade Balance -\$60.3B	fetchers.js
Fix 5	Monthly TIC data replaced Claude estimates with real Treasury slt_table5.txt parsing. Japan \$1,239B, China \$693B, UK \$897B — all verified.	api/tic.js, fetchers.js
Fix 6a	Added "Changes shown are vs prior month (MoM)" label to TIC section	App.jsx
Fix 6b	Added week-ending date to Fed Balance Sheet weekly change field	App.jsx
Fix 6c	Fed Balance Sheet badge now shows actual FRED data date, not today's date	fetchers.js
Fix 7	Intel briefing anchored to verified live prices instead of generic estimates	fetchers.js
Fix 8	WTI/Brent routed from FRED EIA data (accurate, 1-week lag) replacing broken ETF proxies	fetchers.js
Fix 9	WTI/Brent upgraded to near real-time (~30 min) via OilPriceAPI.com. Resolves critical issue where May 6 Iran deal -10% oil drop was not reflected.	api/oil.js, fetchers.js
Fix 10	Cache busting added to all fetchLive() calls (cache:'no-store' + Date.now() timestamp). Cross-platform fix for Safari/Chrome/iPhone/Android stale data.	fetchers.js
Fix 11	Auction status shows PENDING (grey) when Treasury BTC data	api/auctions.js,

	not yet published, replacing misleading STRONG label	App.jsx
Fix 12	Auctions sorted by maturity descending: 30Y → 10Y → 7Y → 5Y → 3Y → 2Y (standard bond market convention)	api/auctions.js
Fix 13	Hidden blank "High yield: %" and "avg undefined%" in auction cards when data not yet available	App.jsx

Data Tier Architecture

Tier	Type	Examples	Refresh Frequency
1	Real-time	Gold, Silver, Mag7, TIC (official), Auction results	Live / same-day
2	Real with lag	WTI/Brent (OilPriceAPI ~30min), VIX/Yields (FRED 1-day), CPI/PCE (monthly release)	30 min to monthly
3	Hardcoded	Fed rate, FOMC dates, Intel price anchors, trade balance, federal debt	Manual update required
4	Pending	True auction tail (IB), Headline DXY ~98, Futures curves	Requires IB account or paid API

Phase 1 API Stack

API	Data Provided	Cost	Key in Vercel
gold-api.com	Gold, Silver spot prices (real-time)	Free tier	VITE_GOLD_API_KEY
Finnhub	Mag7 stocks, SPX/NDX/DJI (ETF proxy), Earnings EPS	Free tier	VITE_FINNHUB_API_KEY
FRED (St. Louis Fed)	Yields, CPI, PCE, GDP, Fed Balance Sheet, VIX, TGA, WTI/Brent historical	Free forever	VITE_FRED_API_KEY
open.er-	EUR/USD, USD/JPY, GBP/USD	Free	None required

api.com	(hourly)		
Treasury Fiscal Data API	Auction BTC, Indirect %, Dealer %, High Yield (same-day)	Free, public	None required
Treasury TIC slt_table5.txt	Major foreign holders — Japan, China, UK etc. (45-day lag)	Free, public	None required
OilPriceAPI.com	WTI and Brent crude near real-time (~30 min)	Free tier	VITE_OILPRICE_API_KEY
Anthropic Claude API	Intel briefing, Fed narrative, Gov report summaries	Paid per token	VITE_ANTHROPIC_API_KEY

PHASE 2 Richer Visualization & Interpretation

© PLANNED – Not yet implemented

Objective

Transform the dashboard from a data display into a macro interpretation tool. Add trend context, composite cycle indicators, and a reference framework so every number can be understood in historical and theoretical context.

Item	Feature	Description	Data Source	Priority
2A	Intel Live Prices	Pass real Daily tab prices (gold, yields, VIX, SPX) into Intel Claude prompt — eliminate all hardcoded anchors. Architecture: pass dailyData as parameter from App.jsx to fetchIntel.	Existing APIs	HIGHEST
2B	Trend Sparklines	12-month inline sparklines for: yields (2Y/10Y/30Y), gold, VIX,	FRED historical	HIGH

		auction BTC/indirect, inflation (CPI/PCE), Fed balance sheet, TGA. Toggle for 6/12/24 months. Color-coded direction.	(free)	
2C	Dalio Deleveraging Meter	Composite 0-100 gauge scoring "Beautiful" vs "Ugly" deleveraging across 4 levers: Austerity, Debt Restructuring, Monetization, Currency Debasement. Green 60-80 = balanced. Red <40 = deflation. Red >80 = inflation runaway.	FRED + Treasury composite	HIGH
2D	Debt Cycle Stage Indicator	Horizontal progress bar showing Stage 1 (Sound Money) through Stage 6 (Normalization) with current position marker. Current US: Stage 3/4 boundary.	Composite derived signals	MEDIUM
2E	Macro Cheat Sheet Panel	In-app collapsible reference panel: Dalio cycle stages, "How Countries Go Broke" warning signs checklist, quick rules of thumb. Sourced from this document.	Static content	MEDIUM
2F	Missing Indicators	Add from FRED (free): 5Y5Y inflation breakeven (T5YIFR), corporate credit spreads (BAMLH0A0HYM2), interest payments/GDP, Bitcoin price as digital gold proxy.	FRED (free)	MEDIUM
2G	EODHD Integration	Subscribe to EODHD (~\$22/month) for: headline DXY ~98 (replaces Fed broad index), direct NDX quote (replaces ETF proxy), 30-year historical data for sparklines.	EODHD ~\$22/month	MEDIUM

Phase 2 Design Principles

- Mobile-first — all additions must work on iPhone PWA
- No breaking changes — each feature added independently and tested before next

- Real data first — no Claude estimates where FRED/Treasury data exists
- One change at a time — no multi-file simultaneous edits
- Verify before deploy — test each API endpoint before wiring to UI

PHASE 3

Investment Guidance & Action Layer

◇ FUTURE – Requires IB account approval

Objective

Turn MacroWatch from a monitoring tool into an actionable decision-support system. Generate investment theses based on Dalio scenario frameworks, provide positioning signals tied to real indicator thresholds, and integrate Interactive Brokers for true auction tail data and futures.

Item	Feature	Description	Dependency
3A	Scenario Framework	5 scenarios tied to Debt Cycle Stage: Muddling Through, Ugly Deflation, Ugly Inflation, Reserve Currency Shift, Recovery/Normalization. Each with indicator conditions and typical positioning.	Phase 2C/2D complete
3B	Dynamic Thesis Generator	Rule-based positioning signals: "VIX >30 + poor auctions + Gold surging = Defensive." Output: traffic light per asset class with 1-sentence rationale anchored to real data.	Phase 2 complete
3C	Wealth Protection Playbook	Updated framework: Core holdings (Gold/Silver/miners, Bitcoin, short-duration TIPS, commodity producers), Hedges (volatility, FX diversification), Avoid/Reduce (long-duration bonds, over-levered equities).	None — can implement now
3D	Interactive Brokers API	True auction tail (when-issued yield), gold/Treasury/oil futures curves, USD/THB, USD/SGD for Asia transition indicators. IB account opened May 3, 2026 — pending approval.	IB account approval

3E	Alert System	Threshold breach notifications for key signals. See table below.	Phase 2 complete
3F	Portfolio Simulator	Hypothetical allocations backtested against historical Dalio-style periods (1970s stagflation, 2008 crisis, 2020 COVID, etc.).	Phase 3D complete

Phase 3 Alert Thresholds

Alert	Threshold	Severity	Dalio Significance
Auction Tail widens	> 2bp	WARNING	Demand shock — buyers demanded higher yield than market expected
Indirect Bidders drop	< 55% for 2 consecutive	CRITICAL	Foreign demand retreat — reserve diversification accelerating
Dealer Takedown surges	> 20%	CRITICAL	No one else wanted it — forced buying, yield pressure ahead
VIX spikes	> 30	WARNING	Fear entering market — late-cycle risk decompressing
Gold breaks key level	> \$5,000/oz	WATCH	Reserve currency confidence loss accelerating
TGA drops sharply	< \$200B	WARNING	Treasury liquidity stress — heavy issuance surge incoming
10Y yield crosses	> 5%	WARNING	Debt service costs escalating — sustainability risk at 122% Debt/GDP
Fed Funds Futures	> 75bps cuts H2	WATCH	Market pricing policy capitulation on inflation
Monthly TIC liquidation	> \$50B net selling	WARNING	Structural foreign reserve reallocation signal

